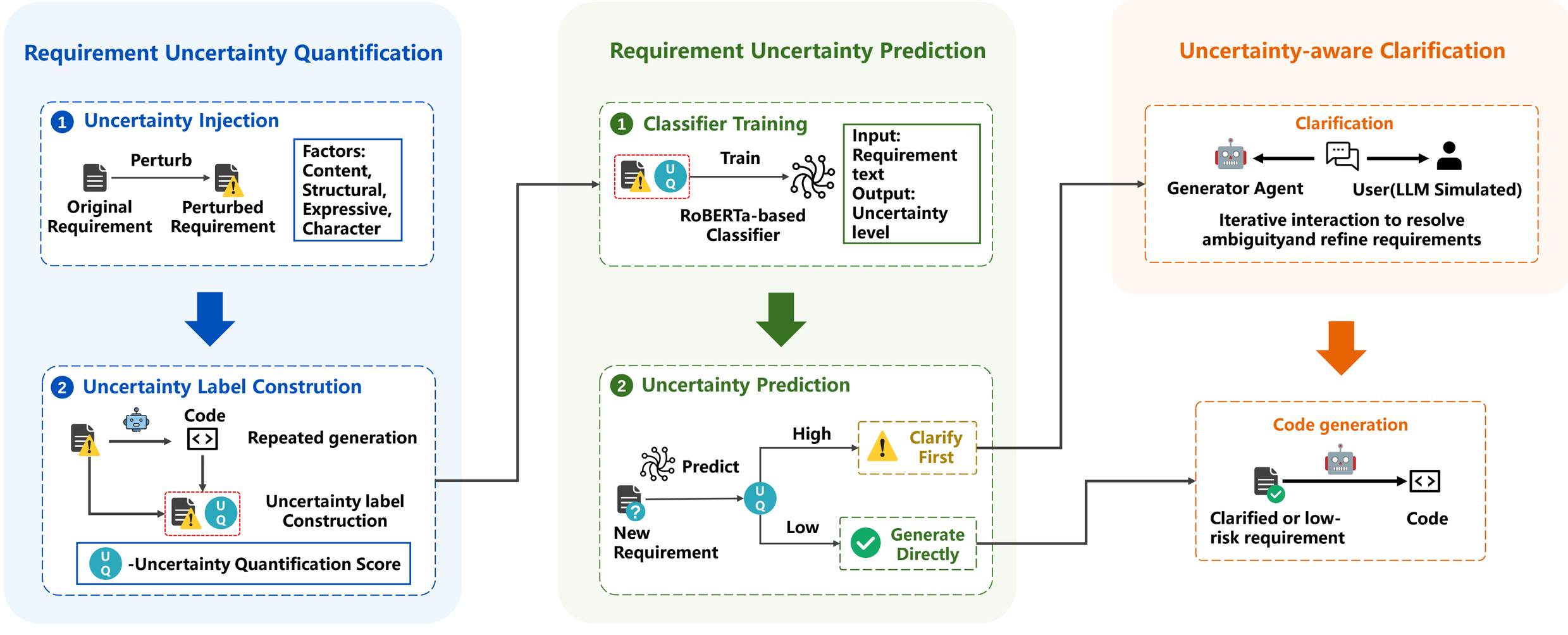

Requirement Uncertainty Quantification
1 Uncertainty Injection
Perturb
Original Requirement
Perturbed Requirement
Factors: Content, Structural, Expressive, Character
2 Uncertainty Label Constrution
Code
Repeated generation
Uncertainty label Construction
-Uncertainty Quantification Score
Requirement Uncertainty Prediction
1 Classifier Training
Train
RoBERTa-based Classifier
Input: Requirement text
Output: Uncertainty level
2 Uncertainty Prediction
Predict
New Requirement
High
Clarify First
Low
Generate Directly
Uncertainty-aware Clarification
Clarification
Generator Agent
User(LLM Simulated)
Iterative interaction to resolve ambiguityand refine requirements
Code generation
Clarified or low-risk requirement
Code